REVIEW

Reconstructing Covert Consciousness

Neural Decoding as a Novel Consciousness Assessment

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Abstract

Determining the level of consciousness in patients with brain injury—and more fundamentally, establishing what they can experience—is ethically and clinically impactful. Patient behaviors may unreliably reflect their level of consciousness: a subset of unresponsive patients demonstrate covert consciousness by willfully modulating their brain activity to commands through fMRI or EEG. However, current paradigms for assessing covert consciousness remain fundamentally limited because they are insensitive, rely on imperfect assumptions of functional neuroanatomy, and do not reflect the spectrum of conscious experience. Neural decoding, in which stimuli and concepts are reconstructed from brain activity, offers a novel approach to covert consciousness assessment that overcomes many of these limitations. In this article, we discuss the current state of covert consciousness assessments, their shortcomings, the state of the science in neural decoding, the potential application of neural decoding to disorders of consciousness, and future directions that may help realize this potential. To do so, we searched PubMed and Google Scholar databases for pertinent articles published between January 1990 and September 2024, using the search terms "covert consciousness," "cognitive motor dissociation," "neural decoding," and "semantic decoding." Redefining covert consciousness with neural decoding may improve sensitivity, enhance granularity, and more directly address the question of what patients can experience after brain injury.

Introduction

For patients with a disorder of consciousness (DoC) after an acute brain injury, discerning their level of consciousness is critical. By level of consciousness, here we refer primarily to awareness or the experience of one's self and environment.[1] This level of consciousness has ethical implications because it may influence how patients are engaged or involved in medical decision making.[2] It also has prognostic implications because patients with higher levels of consciousness in the acute setting are more likely to regain even higher levels of consciousness and function in the future.[3-5] In addition, perhaps most critically, their level of consciousness may influence whether life-sustaining treatments are continued or withdrawn.[6] Thus, answering the frequent question of "what can the patient experience" as accurately as possible is not merely of philosophical interest; it can profoundly affect patient care.

Nonetheless, the objective assessment of consciousness remains challenging. For decades, a detailed examination of behavior, using standardized scales such as the Coma Recovery Scale–Revised,[7] has been the gold standard for discerning a patient's level of consciousness. A surprising finding in 2006 radically altered this landscape, when a patient who appeared unconscious was nonetheless discovered to demonstrate willful modulation of brain activity in response to commands during a fMRI scan. When asked to imagine playing tennis, she demonstrated activity in regions associated with motor planning, and when asked to imagine walking through her house, she demonstrated activity in regions associated with spatial navigation, just as healthy individuals did. This phenomenon, whereby behaviorally unresponsive

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Glossary

BOLD = blood oxygen level dependent; **DoC** = disorder of consciousness; **ECoG** = electrocorticography; **fNIRS** = functional near-infrared spectroscopy.

patients demonstrate evidence of awareness, has since been discovered using similar paradigms in others with DoC[9] and has been termed "covert consciousness"[10] (other terms such as "cognitive motor dissociation" have also been used[11-13]). Covert consciousness has also been identified using EEG, in which patients are instructed to follow a command and a machine learning algorithm,[5,14] or a comparison of spectral densities,[15,16] is used to identify a response in the resulting EEG signal. These fMRI and EEG paradigms have identified evidence of covert consciousness in up to 25% of patients without behavioral responses to commands.[17]

These technological advances demonstrate that patient behavior imperfectly reflects the underlying level of consciousness. As such, the identification of covert consciousness through nonbehavioral detection of command following has attracted increasing attention, warranting inclusion in clinical DoC guidelines across the United States[18] and Europe.[19] However, there remain critical conceptual and practical shortcomings to the current paradigms used to identify covert consciousness. Intriguingly, a growing literature on neural decoding offers a novel approach that may surmount many of these challenges, an application that remains to be tested. In this article, we discuss the limitations of current covert consciousness assessments, outline how neural decoding might be used to redefine our understanding and identification of preserved mental states, and propose future directions that may help realize this potential.

Search Methods

In this narrative review, PubMed and Google Scholar databases were used to identify pertinent articles published between January 1990 and September 2024. Articles were identified using the search terms "covert consciousness," "cognitive motor dissociation," "neural decoding," and "semantic decoding." Representative articles were selected for inclusion based on whether their findings reflected those of other studies in the field, highlighted limitations of current paradigms, and/or illustrated the opportunities or challenges of decoding paradigms.

Limitations of Current Assessments of Covert Consciousness

Technique-Specific Limitations of Current Paradigms

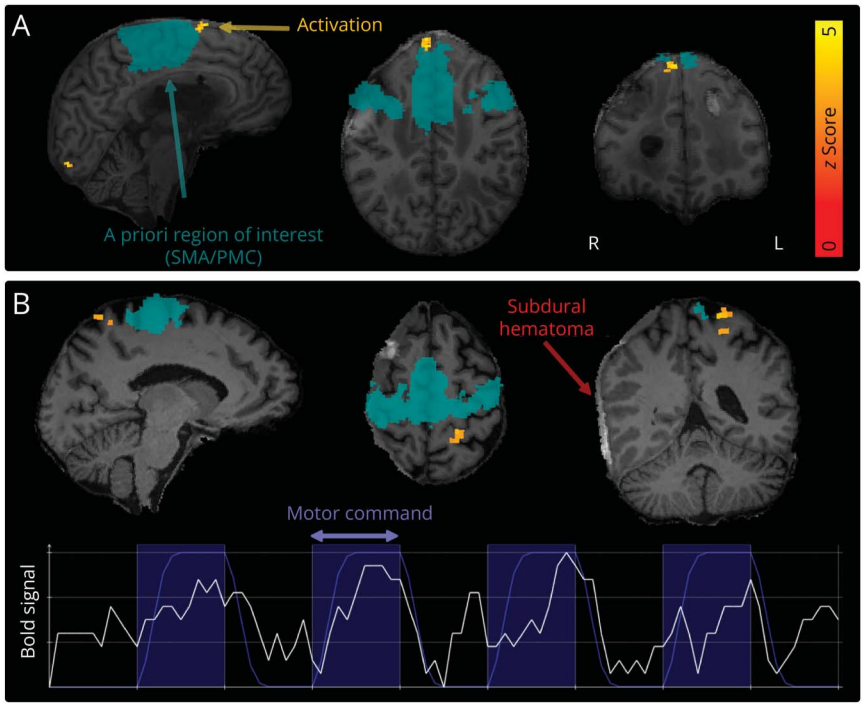
In most current fMRI paradigms, behaviorally unresponsive patients are instructed to perform a task, such as motor or spatial imagery, during an fMRI scan. Motor tasks may be active (e.g., "move your hand") or imagined (e.g., "imagine moving your hand"). Commands are typically interleaved with periods of rest (e.g., "move your hand" and then "stop moving your hand"). The fMRI data are then examined to determine whether changes in neural activity accompany the periods of command delivery (as inferred by changes in the blood oxygen level dependent [BOLD] signal in response to the command relative to periods of rest). This analysis is typically restricted to predefined anatomical regions of interest. For paradigms using motor tasks, neural responses may be expected a priori in regions such as the supplementary motor area and premotor cortex.[20,21] For paradigms using spatial imagery, expected regions may include the parahippocampal gyrus or occipital parietal junction.[8,21-23] If the signal increases beyond a statistical threshold in any of these expected areas in response to the command, patients are concluded to have covert consciousness. In other words, covert consciousness in current fMRI paradigms is typically defined by significant activation in brain regions relevant to the administered task. Limiting analysis to a priori regions reduces the risk of false-positive results, both by reducing the number of measurements across the brain and by excluding regions (e.g., auditory cortices) that may reflexively respond to commands without necessarily reflecting awareness.

There are several shortcomings of this standard approach. First, even in the healthy brain, there is substantial variation in functional anatomy, and thus, the expected location of task-elicited activity is not necessarily clear. For example, in response to motor tasks, healthy individuals may demonstrate activity outside the supplementary motor area or premotor cortex.[20,24] Second, in the setting of acute brain injury, which is often associated with structural distortion of the brain, expected regions may become even more difficult to identify. Third, it is uncertain what statistical threshold is necessary for concluding that brain activity is meaningful. Thresholds that are too lax may lead to the misinterpretation of noise as evidence of covert consciousness (false positives), and thresholds that are too stringent may lead to failure to detect true responses (false negatives). The appropriate threshold is unknown in the absence of a gold-standard consciousness assessment with known and fixed signal and noise characteristics.

We provide examples that illustrate the limitations and uncertainties of the command-based fMRI paradigm (Figure 1). Figure 1A illustrates a scan in which a subset of activation only slightly overlaps the a priori regions of interest (supplementary motor area and premotor cortex). It is uncertain whether any degree of overlap should be considered representative of covert

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Figure 1 Limitations of Current fMRI-Based Convert Consciousness Assessments



Examples of current fMRI-based convert consciousness assessments are provided. Scans were accompanied by instructions to "imagine moving your right hand" and then to stop, for 17.5-second blocks, for a total of 2 minutes and 20 seconds. Data were processed using motion correction, smoothing with a Gaussian kernel of 4-mm full width at half maximum, and high-pass temporal filtering with a cutoff of 35 seconds. For statistical analysis, a design matrix comprising a single explanatory variable was modeled using a square wave function convolved with a gamma hemodynamic response function, with temporal derivatives included to account for the shift in response timing. Confound regressors, including motion parameters and physiologic regressors, were incorporated into the model for artifact correction. A single contrast was defined to compare periods of commands with periods of rest. The resulting statistical maps were thresholded using cluster-based correction, with a Z threshold of 3.1 and a corrected cluster significance threshold of $p < 0.05$. The regions of significant activation were superimposed onto an a priori region of interest comprising the supplementary motor area and premotor cortex (SMA/PMC). In scan A, the cluster of activation only slightly overlapped the region of interest. In scan B, a subdural fluid collection potentially distorted typical functional neuroanatomy. The cluster of activation did not overlap the region of interest, but the BOLD signal within the region of activation (white tracing) showed consistent elevations during periods of command (blue shaded periods, with the explanatory variable model represented by blue tracing), compared with periods of rest. BOLD = blood oxygen level dependent.

consciousness, particularly in cases where, despite rigorous statistical thresholding, activation is distributed across several brain regions. Figure 1B illustrates a scan with structural distortion of the brain due to a subdural hematoma. The scan demonstrates a reliable response outside the a priori regions. It is uncertain whether such activation outside a priori regions could also represent covert consciousness.

Similarly, in the current EEG paradigm to assess covert consciousness, behaviorally unresponsive patients are typically instructed to perform a motor task, interleaved with periods of rest. Machine learning, or a statistical comparison of EEG frequencies (i.e., spectral densities), is used to determine whether tasks elicit systematic changes in the distributed responses across electrodes reflective of covert consciousness.[5,14-16] In other words, covert consciousness in current EEG paradigms is typically defined by a change in brain activity that accompanies an administered task. The EEG paradigm overcomes some shortcomings of the fMRI paradigm, and combining the 2 increases the sensitivity of covert consciousness detection.[17] Because the EEG paradigm is agnostic to brain regions, it is less susceptible to error in cases of atypical functional topography or structural brain distortion. However, independence from neuroanatomy carries its own limitations. Because the command-specific EEG changes could occur anywhere throughout the brain, it is difficult to be certain that such changes reflect an attempt to execute the command. It is possible, for example, that the difference in EEG signal between a command (e.g., "keep moving your hand") and a rest period (e.g., "stop moving your hand") could reflect differences in the auditory quality of the stimulus, rather than an attempt to execute the command per se.[14] Supporting that possibility, 1 study using this paradigm identified conscious patients with intracerebral hemorrhage and global aphasia who demonstrated, despite overtly impaired language comprehension, EEG responses to commands.[25]

Fundamental Limitations of Current Paradigms

Apart from the shortcomings specific to each technique, there are also more fundamental limitations that generalize across current covert consciousness assessments. First, even in overtly conscious individuals, these techniques have imperfect sensitivity for detecting consciousness. fMRI-based and EEG-based techniques have missed evidence of consciousness in up to 25% of healthy individuals[20,26] and 60% of brain-injured patients with behavioral responses to commands.[17,25] Such insensitivity may reflect functional neuroanatomical variants, fluctuations in arousal or attention, inappropriate statistical thresholding, artifact, or other technical failures.

Second, these paradigms are inherently insensitive for detecting consciousness in patients with brain injury. For a patient to demonstrate covert consciousness, they must attend to the stimulus (precluded by delirium or lesions that deplete attentional resources, such as basal forebrain contusions), comprehend the command (precluded by receptive

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aphasia), translate that command into a coherent response (precluded by poor executive function), have the motivation to execute that response (precluded by abulia), and must continue to do so over repeated trials. Many of these precluding conditions are common in the setting of brain injury (due to the brain injury itself, sedatives, or toxic-metabolic insults), and any one may be sufficient to obscure the detection of consciousness.

Third, these paradigms treat consciousness as a dichotomous phenomenon. Consciousness exists along a spectrum, ranging from the most basic perception of sounds and sights, to the discernment of objects, to the understanding of concepts, and finally to the ability to interact. Current paradigms for covert consciousness simplistically binarize this spectrum—covert consciousness may either be present or absent—while setting a high bar for the demonstration of consciousness.

Fourth, current paradigms do not facilitate the development of brain-computer interfaces. Perhaps one of the most important implications of covert consciousness is the possibility of re-establishing communication. To date, however, the paradigms used for the detection of covert consciousness have generally not been translated into a medium of communication. As illustrated by 1 case reported in 2010[22] and 2 cases reported in 2013,[27] rare patients can use the fMRI paradigm to answer yes-or-no questions, by imagining either a motor or spatial task, respectively, but this method demands significant effort and capability from the patient and cannot convey more complex concepts.

Ultimately, the current paradigms of covert consciousness assessment are limited in answering the fundamental question, "What can the patient experience?" The absence of covert consciousness on command-based paradigms does not necessarily indicate that a patient lacks experience—the absence of evidence is not the evidence of absence (i.e., the absence of command following does not preclude the possibility of awareness).[28] Similarly, given their risk of false positives, a positive result of current paradigms may not necessarily assure the presence of normal conscious experience. Current paradigms are neither necessary nor sufficient in determining a level of consciousness.

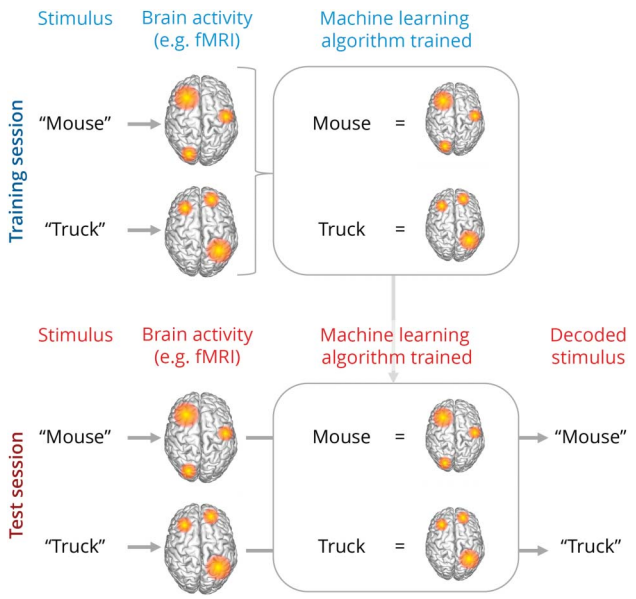
In addition to the "task-based" paradigms discussed, there are numerous other "stimulus-based" methods that have been developed for probing brain function after brain injury.[29] For example, both fMRI[23,30,31] and EEG[32-34] have been used to identify cortical responses to sensory stimuli in behaviorally unresponsive patients. Similarly, transcranial magnetic stimulation has been used to evoke EEG responses; the complexity of this response has been correlated with level of consciousness.[35-37] These techniques help to address the uncertain neurologic function in patients with negative results of current covert consciousness assessments. However, it is also uncertain whether the passive response of the brain to stimulation (covert cortical processing[38]) is sufficient to indicate the presence of conscious experience.[28]

There is thus a critical need for a new approach to defining and identifying covert consciousness, one that overcomes the conceptual and practical shortcomings of current paradigms. An ideal method for probing consciousness, relative to current paradigms, would enhance sensitivity without sacrificing assurance of awareness, would depend less on prespecified regions of interest, would require less intensive patient participation, would be sensitive to gradations in levels of awareness, would facilitate the development of brain-computer interfaces, and ultimately, would more directly address the question of what patients experience.

Neural Decoding

An ideal scenario would be to "decode," or reconstruct, the conscious experience of a patient based on their neural activity. Though once the ambitions of science fiction, neural decoding has become increasingly feasible with advances in technology. The principles of neural decoding are typically as follows (Figure 2). During a training session, an individual is exposed to a series of sensory stimuli while their resulting

Figure 2 Principles of Neural Decoding



The typical sequence of procedures in neural decoding is represented, using semantic decoding and fMRI as examples (with orange regions representing patterns of brain activity). In a training session, a series of stimuli are presented while a patient undergoes brain activity recording. This data set—including both the presented stimuli and the resulting brain activity—is used to train a machine learning algorithm, which identifies associations between specific stimuli and specific patterns of brain activity. Subsequently, in a test session, a series of stimuli are again presented while a patient undergoes brain activity recording. The previously trained machine learning algorithm is then presented exclusively with the brain activity data set, using the previously established associations to estimate which stimuli produced each pattern of brain activity.

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brain activity is monitored through modalities such as fMRI and EEG.[39] This "training" data set allows a machine learning (or "pattern classification") algorithm to establish associations between stimuli and neural responses. During a subsequent test session, the individual is exposed to a new set of stimuli, and the machine learning decoding algorithm is applied to the associated brain activity to reconstruct the stimuli that the individual had experienced.

This technique can be applied across a spectrum of inputs, from simple auditory and visual stimuli to complex concepts. At a basic level, neural decoding can be used to reconstruct modality-specific sensory experiences (i.e., sensory decoding). More recently, advances in neural decoding have permitted the reconstruction of modality-independent concepts (i.e., semantic decoding), initially distinguishing between broad semantic categories and more recently reconstructing externally or internally derived narratives.

Neural decoding has been used to reconstruct a range of visual stimuli.[40] In 2005, a study in 4 healthy individuals demonstrated that the orientation of a viewed line could be reliably decoded using fMRI.[41] Decoding was modulated by attention; if individuals were presented with overlapping lines, the line to which attention was directed dominated the decoded brain activity. In 2009, a study in 3 healthy individuals demonstrated that images could be decoded using fMRI and a Bayesian decoding approach.[42] In 2011, a study in 3 healthy individuals demonstrated that movies could be decoded using fMRI and a model that allowed rapidly changing stimuli to be decoded from the slower BOLD signals.[43] In 2017, a study in 5 healthy individuals demonstrated that even imagined visual stimuli could be decoded using fMRI.[44]

A similar technique has been used to decode auditory stimuli. In 2011, a study using fMRI in 4 blind individuals demonstrated that the direction of sound movement could be decoded using fMRI.[45] In 2012, a study decoded individual words using intracranial electrodes (electrocorticography [ECoG]) in the auditory cortices of 15 individuals undergoing neurosurgery.[46] In 2019, a study using ECoG in 5 individuals demonstrated that more complex features of language could be decoded, such as the sex of the speaker.[47] Like visual decoding, auditory decoding is modulated by attention; when 10 healthy individuals were asked to attend to 1 of 4 speakers, the attended speech was more accurately decoded with EEG.[48] In 2023, a study of 29 individuals with ECoG demonstrated that a song could be reconstructed.[49]

While these studies have focused on decoding the sensory quality of stimuli, others have made progress in decoding their semantic content.[39] Although fMRI has been the most common technique for doing so, studies have also used EEG, ECoG, magnetoencephalography, functional near-infrared spectroscopy (fNIRS), and other techniques.[39] EEG studies in 2010[50] and 2011[51] demonstrated that semantic categories of stimuli, such as animals or tools, could be distinguished with a mean accuracy between 72% and 79%. In 2023, an fMRI study led by Tang et al.[52] in 7 healthy individuals used a novel decoding model to reconstruct the semantic content of perceived speech, imagined speech, and silent movies. As for sensory decoding, semantic decoding was modulated by attention: when individuals were presented with simultaneous speech, the semantic content from the attended speech was reconstructed with greater fidelity.

Neural Decoding as a Covert Consciousness Assessment

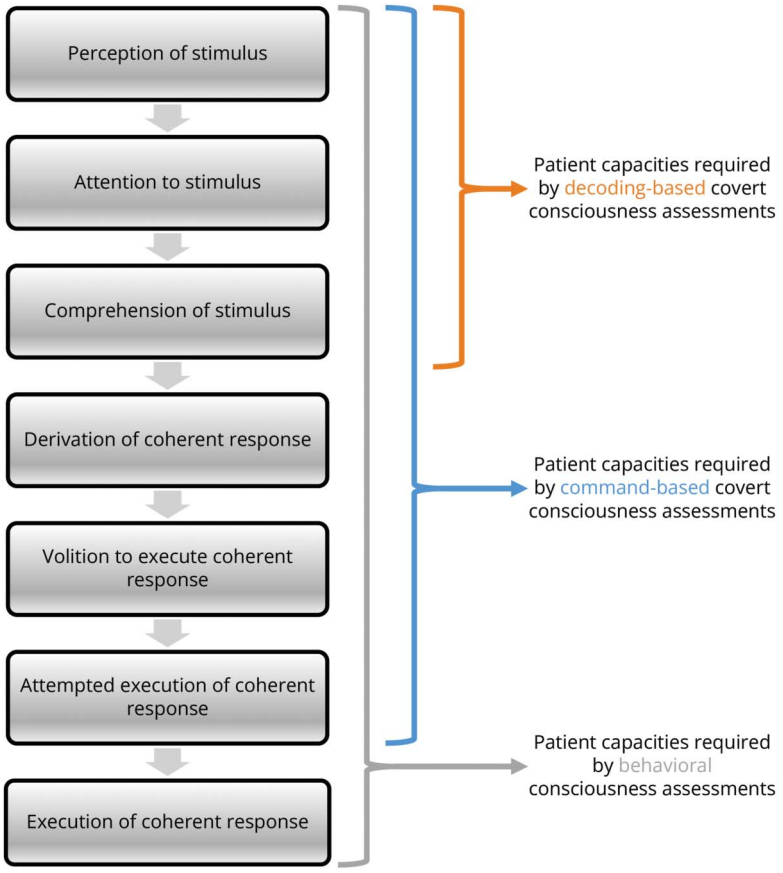
What can neural decoding tell us about the conscious experience of a patient? Next, we will discuss the implications of sensory decoding and semantic decoding and what these may reflect about the spectrum of consciousness. The ability to accurately decode a stimulus implies that the patient's brain activity was sufficiently distinct (i.e., distinguishable from alternative brain states) and reproducible (i.e., reliably associated with the stimulus) to allow the stimulus reconstruction to occur.

Sensory decoding—that is, decoding the qualitative features of a stimulus, such as the sound of language—may not necessarily imply that a patient can experience those features. For example, when healthy individuals are anesthetized and presented with audible words, there is significant activation of the auditory cortices, although they do not recall hearing those words.[53,54] Similarly, in anesthetized patients with postanoxic coma, the distinction between standard and deviant sounds can be decoded with EEG.[55] It is unclear whether any technique, in principle, can differentiate between neural responses that reflect subconscious processing and conscious experience, of sensory stimuli.[28]

However, semantic decoding has stronger implications. Brain activity that is distinctly and reproducibly associated with a *semantic concept* implies that the brain can process a stimulus to the point of *extracting meaning*. The extraction of meaning more convincingly implies the presence of consciousness. It is arguably the extraction of meaning—the experience of the world not only as a jumble of sensory stimuli but as a set of discrete objects and concepts—which fundamentally constitutes awareness of one's self and environment. Moreover, the existing paradigms for assessing covert consciousness have already established the extraction of meaning as a benchmark for consciousness. The demonstration of command following, as assessed by current paradigms, is accepted to indicate consciousness not because of the attempted execution of the commanded behavior per se (which reflects volition, above and beyond mere consciousness) but because the execution of the commanded behavior indicates that the command itself was comprehended. Or, in other words, command following compellingly indicates consciousness because it implies that patients extract meaning from the command. Semantic decoding establishes that the extraction of meaning has

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Figure 3 Capacities Required by Different Consciousness Assessments

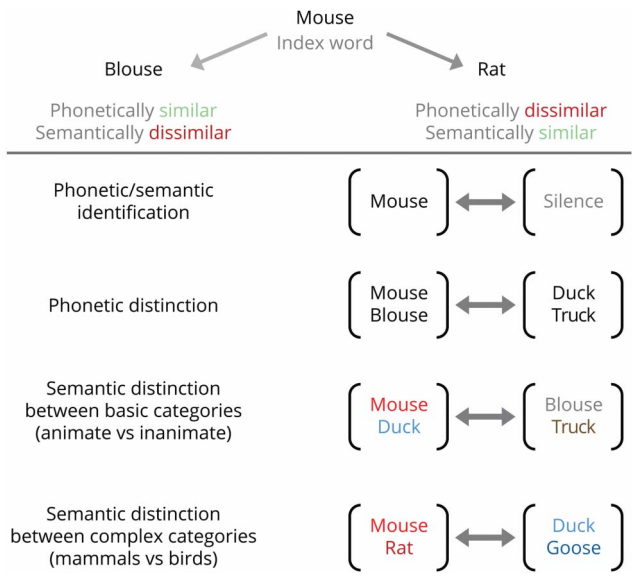


Traditional behavioral consciousness assessments require a series of intact capacities for conscious patients to be detected as such. Command-based covert consciousness assessments eliminate the necessity for behavioral execution but continue to require most of those same capacities. Decoding-based covert consciousness assessments eliminate the reliance on many of those capacities for signs of consciousness to be identified.

occurred without relying on the attempted execution of a response (Figure 3).

Neural decoding permits not only the inference of consciousness but also discernment between different levels of consciousness. Current command-based consciousness paradigms are dichotomous and only determine whether patients do or do not have sufficient semantic extraction to comprehend the meaning of a command (assuming that they also have the necessary capacities to attempt its execution). By contrast, decoding paradigms permit the assessment of consciousness along its spectrum, evaluating multiple levels of complexity (Figure 4). For example, if a patient is presented with a series of words describing birds, mammals, tools, and vehicles, one can determine the extent to which the patient can distinguish phonetically dissimilar words, animate from inanimate objects (a basic semantic distinction), or birds from mammals (a more complex semantic distinction). The extent to which a patient can make such distinctions, at each level, will be reflected by the *fidelity* with which reconstructed stimuli and concepts resemble those presented. A high fidelity of semantic reconstruction (which can be quantified with metrics such as the BERTScore)[52] implies a high degree of distinct and reproducible brain states to those semantic categories and thus a high degree of extraction of meaning.

Figure 4 Hierarchy of Decoding



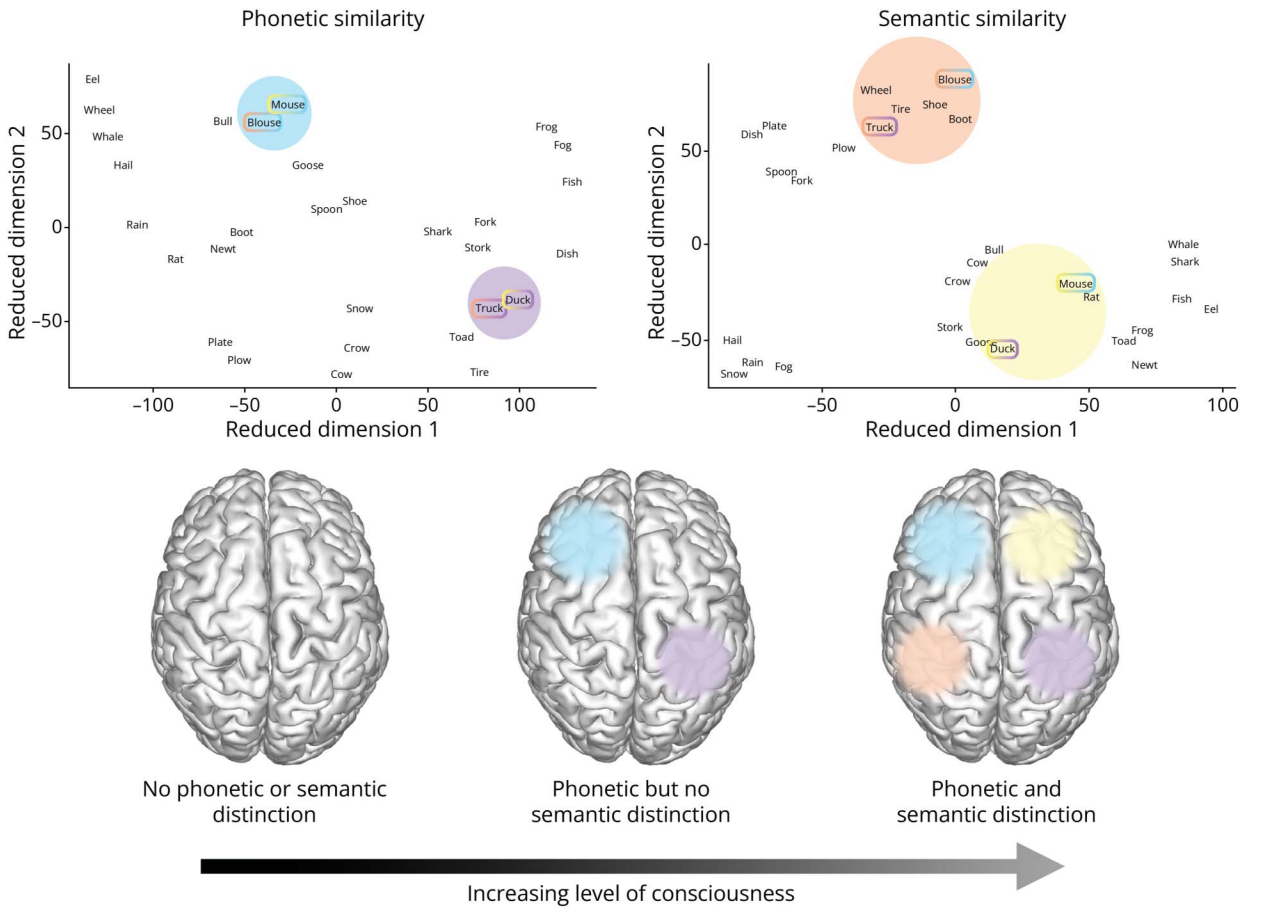
The spectrum of consciousness can be evaluated by assessing the brain's capacity to distinguish between different kinds of stimuli. For any word, analogs that share either phonetic or semantic similarity (and are otherwise matched for features such as syllables) can be used to assess increasing levels of sensory and conceptual complexity.

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Furthermore, levels of distinction need not be organized categorically but can be organized continuously, at both sensory and semantic levels (Figure 5). For example, semantically, a mouse is only slightly different from a rat, more different from a bull, more different still from a truck, and so on. Phonetic and semantic differences between words can be conceived as continuous "distances," or embedding vectors, using one of several methods. For example, phonetic embeddings can be retrieved from a model using a vowel-weighting method derived from the Carnegie Mellon University Pronouncing Dictionary,[56] and semantic embeddings may be retrieved from OpenAI's text-embedding-3-large model (January 25, 2024). Pairwise cosine similarities, for example, can then be computed between each word pair. The degree to which a patient's brain can discern increasingly short phonetic and semantic distances (higher cosine similarities), as indicated by an increasing fidelity with which stimuli can be decoded, will reflect the degree to which the patient can process stimuli and extract their meaning. In other words, patients with higher levels of consciousness may be capable of distinguishing between more similar stimuli; for example, a more cognitively intact patient may be capable of discerning a mouse from a rat while a less cognitively intact patient may only be capable of discerning a mouse from a truck. There is unlikely to be any specific fidelity threshold beyond which consciousness can be presumed and below which consciousness can be presumed absent. However, in comparison with current paradigms, these metrics may more faithfully reflect the continuous range of possible states of consciousness and more directly address what a patient can experience.

Another advantage of decoding-based paradigms, compared with current paradigms that require a priori regions of interest, is that they are more data driven and rely purely on patient-specific models, thereby reducing error caused by atypical functional topography or structural distortion (Table). Although decoding paradigms may exclude brain regions that are less likely to demonstrate stimulus-specific activity (e.g., white matter), they are, compared with current

Figure 5 Continuum of Decoding



Phonetic and semantic distinctions can be organized continuously, by conceiving phonetic and semantic "distances," or embedding vectors, between words. Here, phonetic embeddings are based on a model developed derived from the Carnegie Mellon University Pronouncing Dictionary,[56] and semantic embeddings are based on OpenAI's text-embedding-3-large model (January 25, 2024). Pairwise cosine similarities can then be computed between each word pair. For visualization purposes, a 2-dimensional t-distributed stochastic neighbor embedding (t-SNE) analysis was performed to reduce the dimensionality to 2 dimensions.[59] This was run using the Rtsne package (version 0.17) with the same arguments for both plots (dims = 2, perplexity = 4, theta = 0, and a fixed random seed of 123). The brain's capacity to discern increasingly similar phonetic and semantic distinctions may reflect an increasing level of consciousness.

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Table Limitations of Current Covert Consciousness Paradigms and Potential Solutions of Neural Decoding

	Limitations of current paradigms	Potential solutions of neural decoding
fMRI	Dependent on a priori regions of interest, presumption of conserved functional neuroanatomy	Less dependent on a priori regions of interest, tailored to individuals
	Dependent on arbitrary statistical thresholding	Not dependent on arbitrary statistical thresholding
EEG	Uncertain reflection of sensory or semantic content of stimulus	Assesses sensory and semantic content separately
Fundamental	Dependent on patient cooperation	Less dependent on patient cooperation
	Covert consciousness established as a dichotomous phenomenon	Covert consciousness assessed along a spectrum
	Difficult to translate into a brain-computer interface	Amenable to translation into a brain-computer interface
	Indirect assessment of patient experience	More direct assessment of patient experience

fMRI paradigms, relatively agnostic as to where brain activity occurs. As long as the elicited brain activity is distinct and reproducible, even if it varies drastically in spatial distribution between patients, it can be decoded to reflect the patient's level of consciousness.

Limitations to Decoding Paradigms

There are limitations to covert consciousness assessments that are not resolved by decoding paradigms. As discussed, this approach does not avoid the inherent uncertainty in evaluating a fundamentally subjective state. Could the extraction of meaning occur "reflexively"? Is it possible that a brain could demonstrate distinct and reproducible activity to semantic concepts, without the individual actually experiencing those concepts? Convolutional large language models may be argued to extract concepts from language without ever consciously experiencing those concepts. Ultimately, reflexive and conscious brain activity may never, in principle, be confidently distinguished objectively.[28] Even command following may be achieved by robots without reflecting true experience. However, semantic extraction, which produces the concepts of self and environment that constitute our very definition of consciousness, offers a reasonable threshold for inferring consciousness.

There are additional limitations in how decoding is interpreted and conveyed. As for current paradigms, a negative result (in the case of decoding, a low reconstruction fidelity) does not necessarily imply the absence of consciousness. Although decoding paradigms will avoid certain causes of false-negative results (e.g., avolition), they remain susceptible to others, such as inattention, deafness, aphasia, or technical artifact. Notably, even these remaining causes are potentially more surmountable in decoding paradigms than in current paradigms—attention may be more easily drawn to a salient object than to a command, and deafness or aphasia may be circumvented through alternative stimulus modalities (the feasibility of semantic decoding from visual stimuli has also been established).[52] In addition, there are limitations specific to decoding paradigms as a means of assessing consciousness. For example, the continuous output of this technique complicates interpretation; although it may more faithfully represent the continuous spectrum of consciousness, it may be difficult to concretely convey its results to clinical teams or family members.

There are also technical hurdles to the clinical implementation of decoding paradigms. Previous studies have acquired extended durations of data (several hours) to train machine learning algorithms, with evidence that decoding accuracy increases with longer durations.[52] It is possible that factors such as inattention or heterogeneous neural responses to stimuli could mandate extended data collection in patients with brain injury as well. This could present a challenge, given that hours of data acquisition may not be feasible in this critically ill population. However, decoding paradigms may be achievable in shorter time spans. First, accurate semantic decoding in healthy individuals, at levels significantly higher than chance, could be achieved after approximately 1 hour of auditory stimulus delivery and after even shorter durations of visual stimulus delivery.[52] Moreover, for the purposes of consciousness assessment, reconstructing full narratives is unnecessary; reconstructing basic concepts should be sufficient, which may be achievable with less data. Finally, decoding is less accurate with more abstract concepts[52]; presenting more concrete concepts may help reduce the data required to train a machine learning algorithm.

Ethical Considerations

A concern for mental privacy has accompanied the development of neural decoding technology. If decoding paradigms become sufficiently advanced to permit the reconstruction of perceived or imagined concepts, it may be possible to expose an individual's thoughts against their will. Arguments against this concern have included that it is (1) practically difficult to involuntarily restrain someone for long enough to acquire the data necessary for neural decoding and (2) possible for individuals to consciously resist the reconstruction of their thoughts by willfully engaging in distracting tasks.[52] These arguments, however, may not apply to patients with DoC, from whom data can be acquired

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without cooperation. Moreover, they may be too encephalopathic to recognize the paradigm, independently devise a distracting task, and consistently engage in that task. It, therefore, must be acknowledged that the prospect of semantic decoding in patients with DoC risks exposing their mental content without their explicit consent, which itself is impossible to obtain.

There are several reasons to believe that semantic decoding may nonetheless be ethically justifiable in patients with DoC. First, by design, the decoding paradigm seeks to reconstruct stimuli that are delivered to the patient, not internally derived stimuli. Second, even if patients fail to attend to the delivered stimuli, attending instead to their inner thoughts, the machine learning algorithms will not have been trained to decode those inner thoughts. Third, additional safety precautions may mitigate the risk of decoding semantic content against the patient's will. At the start of both the training and test data sets, a statement like the following can be included in the instructions to patients: "We are conducting a study to better understand your level of consciousness. To do so, we will attempt to reconstruct the stimuli that you are about to experience. If you do not want us to reconstruct those stimuli, please repeat to yourself 'stop' instead of paying attention to the stimuli." Leveraging the distractibility of semantic decoding paradigms, this strategy may help patients (at least those with a high level of covert consciousness) to willfully obscure their mental content.

There nonetheless remains an unavoidable risk, no matter how unlikely or limited, of exposing a patient's mental content against their will. However, it must be recalled that understanding a patient's level of consciousness and capacity for experience may dictate whether life-sustaining treatments are continued or withdrawn. It is therefore possible that, at least in some circumstances, the potential benefits of more accurately assessing a patient's level of consciousness may outweigh the potential risks of involuntary semantic decoding. Developing and performing such testing in multidisciplinary collaboration with an ethicist,[57] who can help evaluate the risks and benefits and assist in the interpretation or communication of results, will be critical.

Conclusion and Future Directions

Throughout history, the brain has been understood in terms of the most current technology. As technology has evolved, so too have our metaphors for the brain—as clockwork, as a telephone switchboard, and ultimately as a digital computer. As we transition to a new era of machine learning and artificial intelligence, it seems fitting that our conception of consciousness would make a similar transition. We accept that the exact processes by which artificial intelligence operates may never be completely understood to us, and yet, we acknowledge its pragmatic utility in helping us to understand our world and make decisions. Similarly, while the precise neural mechanisms of consciousness may never be known to us, we can use artificial intelligence to make inferences about a person's level of consciousness. Although current covert consciousness assessments have been easily conceptualizable, they remain fundamentally insensitive, oversimplifying consciousness into a binary state, with results that are therefore predictably often ambiguous. To more directly answer the question of what a patient with DoC can experience, we may need to abandon the comfort of easily conceivable paradigms and trust artificial intelligence to reveal a broader spectrum of consciousness.

Despite the promises of neural decoding as a novel means of evaluating consciousness, there are undoubtedly many practical hurdles that must be overcome before this promise is realized. Feasibility of this approach will require acquiring data of sufficient spatial and temporal resolution, acquiring the necessary duration of data, implementing the techniques required for data analysis, and appropriately navigating the ethical complexities, all in critically ill patients with brain injury. Accessibility of such technologies poses another challenge. It is notable that even with reduced spatial resolution, matching the resolution of more portable technologies such as fNIRS, neural decoding has still been demonstrated to be feasible,[52] suggesting that more portable alternatives may be possible for the future (although such alternatives may be more susceptible to artifact).

If neural decoding is demonstrated to be feasible as a covert consciousness assessment, there are many further avenues for investigation. Questions relevant to clinical implementation include whether certain etiologies of brain injury are more amenable to neural decoding than others and whether test performance varies based on stimulus modality (e.g., auditory vs visual). Critical questions will be whether decoding fidelity predicts future recovery, as current covert consciousness paradigms do,[5,14] and whether semantic decoding can be effectively translated to a brain-computer interface. A brain-computer interface would typically require signal acquisition, signal preprocessing, feature extraction, and translation to an external device.[58] Conveniently, neural decoding would already account for the first 3 steps, although the repertoire of concepts used to train the machine learning algorithm would need to be expanded to account for the possible range of internally derived concepts. The machine learning algorithm could then be used not merely to reconstruct external stimuli but to decode the patient's internal concepts, which could be transmitted through a communication device (e.g., screen or speaker). If such a system proves feasible, there will be additional technical and ethical considerations. Will it be possible to discriminate between concepts the patient wishes to communicate and those they do not? And, can a brain-computer interface achieve the reliability and complexity necessary to evaluate a patient's decision-making capacity, and if sufficient, to incorporate them in medical decision making?

Despite the hurdles and uncertainties ahead, the fundamental limitations of current approaches and the theoretical promises

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of neural decoding suggest that this is an important path of investigation in the evaluation of consciousness in DoC.

Author Contributions

D. Fischer: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. B.L. Edlow: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. H.J. Freeman: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. D. Alaiev: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. Q. Wu: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. J.B. Ware: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. J.A. Detre: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. G.K. Aguirre: drafting/revision of the manuscript for content, including medical writing for content; study concept or design.

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